

CA5 Question 2 Report

TravBot Live Agent System

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1 Summary

This report documents the final Q2 implementation of TravBot as a strict live-integration multi-tool travel agent. The system integrates OpenAI, Amadeus, Tavily, Kaggle datasets, LlamaParse, LanceDB, and LangGraph. The code is modularized under `answers/Q2/src` and produces reproducible scenario logs and report plots.

2 Implementation Outputs

Artifact	Description
<code>answers/Q2/lancedb_data/</code>	Vector database for FAQ and tourism content
<code>answers/Q2/results/scenario_results.csv</code>	Scenario-level outputs, latency, and tool usage
<code>answers/Q2/results/scenario_results.json</code>	JSON export of scenario outputs
<code>answers/reports/figures/q2/*.png</code>	Report-ready scenario and performance plots

3 System Architecture

The Q2 system follows this deterministic sequence:

1. Runtime key loading from environment files
2. Live preflight checks for tools, modules, and credentials
3. Kaggle dataset ingestion for IATA and currency mappings
4. LlamaParse extraction from the tourism PDF
5. LanceDB initialization for FAQ and tourism retrieval
6. LangGraph ReAct agent construction with seven tools

7. Scenario execution and output logging
8. Plot generation for report publication

4 Tooling Coverage

The agent uses the following tools:

- Flight search via Amadeus
- Hotel search via Amadeus
- Restaurant search via Tavily
- Weather information via Tavily
- Currency conversion context via Tavily and Kaggle mappings
- FAQ semantic search via LanceDB
- Trip planning synthesis using OpenAI and retrieved contexts

5 Figures

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Figure 1: Scenario latency by scenario ID

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Figure 2: Latency distribution

Missing figure: figures/q2/q2_tool_usage_frequency.png

Figure 3: Tool usage frequency

Missing figure: figures/q2/q2_query_length_vs_latency.png

Figure 4: Query length vs latency

6 Reproducible Execution

All commands are executed under:

```
source /Users/tahamajs/Documents/uni/venv/bin/activate
```

Primary command:

```
bash answers/scripts/run_all.sh
```

7 Validation Criteria

The final implementation targets the following checks:

- Strict live preflight passes with required credentials and services.
- Notebook execution completes from top to bottom.
- Scenario output files are generated and non-empty.
- All Q2 figures are generated under `answers/reports/figures/q2`.
- Report builds succeed with XeLaTeX into `answers/reports/build`.